

Communication Laboratory

Experiment 3: AM MODULATION USING IC MC 1496



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1. Introduction

Amplitude Modulation (AM) is a widely used modulation technique in communication systems where the amplitude of a high-frequency carrier signal is varied in proportion to the instantaneous value of the message signal, while the carrier frequency and phase remain constant. In a conventional AM signal, the transmitted wave consists of the carrier component along with two sidebands, known as the upper sideband (USB) and lower sideband (LSB), which contain the information of the modulating signal.

However, in standard AM transmission, a large portion of power is carried by the carrier which does not contain useful information, making the system power inefficient. To overcome this drawback, Double Sideband Suppressed Carrier (DSB-SC) modulation is used, where the carrier component is suppressed and only the two sidebands are transmitted. This improves power efficiency as the transmitted power is concentrated in the sidebands that carry the information.

In this experiment, a balanced modulator using the MC1496 IC is implemented to generate a DSB-SC signal. The IC contains differential amplifier stages that enable switching with respect to the carrier signal and amplification of the message signal. The experiment also involves observing different conditions of modulation such as under-modulation, critical modulation, and over-modulation by varying the modulation index, and analysing the resulting time-domain waveform and frequency spectrum.

2. Objectives

Task 1: Designing circuit for amplitude modulation using the IC MC 1496, making connections as shown:

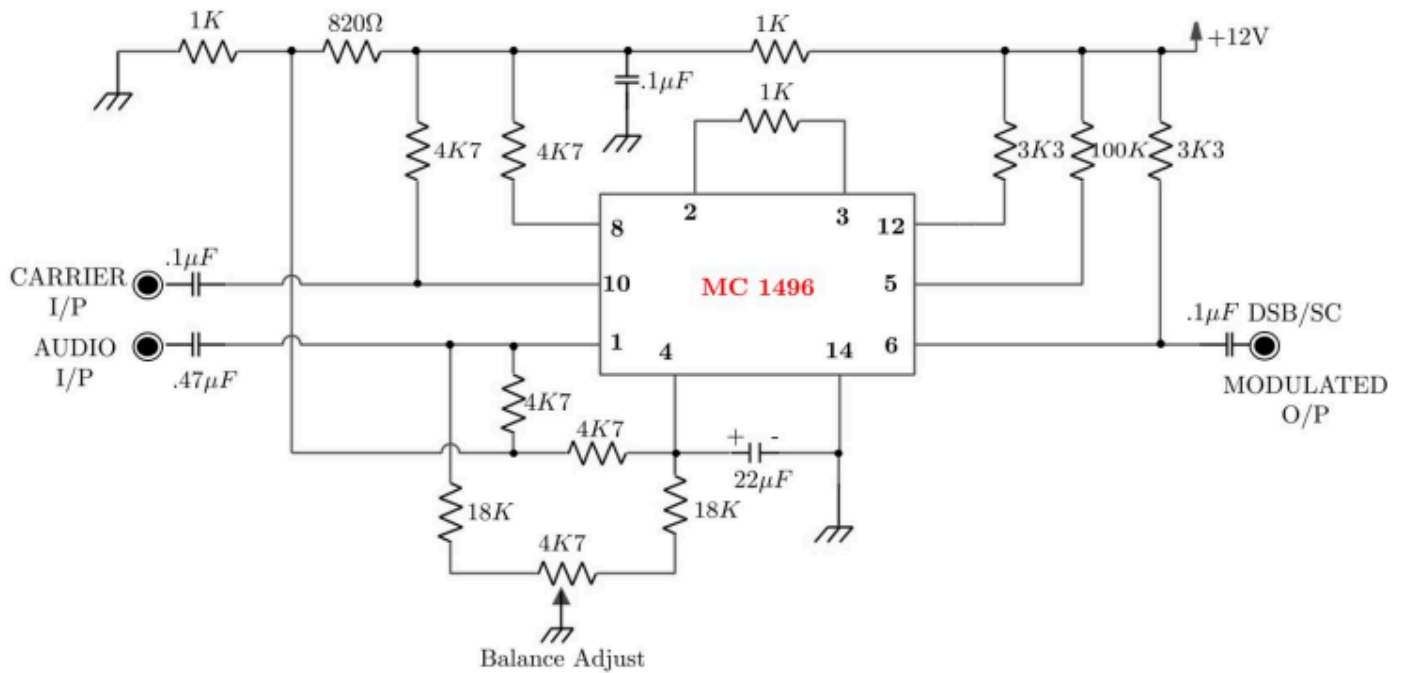


Figure 1: Circuit Diagram

Task 2: Set the values of the carrier and modulating signals as sinusoidal signals with as $A_c = 300\text{mV pp}$, $f_c = 20\text{kHz}$ and $A_m = 150\text{mV pp}$, $f_m = 2\text{kHz}$.

- Observe the modulated output at (Pin No. 6) and adjust the potentiometer (Variable Resistor - Balance Adjust) so that the signal at AM output is maximum without distortion.
- Calculate the Modulation Index (m) using Input Modulating Signal Amplitude and Carrier Signal Amplitude, A_m and A_c respectively as:

$$m = \frac{A_m}{A_c}$$

Or use the AM Modulated Output Waveform Characteristics, $A_{\max}$ and $A_{\min}$ as:

$$m = \frac{A_{\max} - A_{\min}}{A_{\max} + A_{\min}}$$

- Observe the Modulated Output Signal in frequency domain and find f_c , $f_c + f_m$, $f_c - f_m$

Task 3: Vary the amplitude of the message signal A_m and observe the modulated output waveform in both time domain and frequency domain.

Based on the modulation index (m), three cases are observed:

- Under Modulation ($m < 1$)

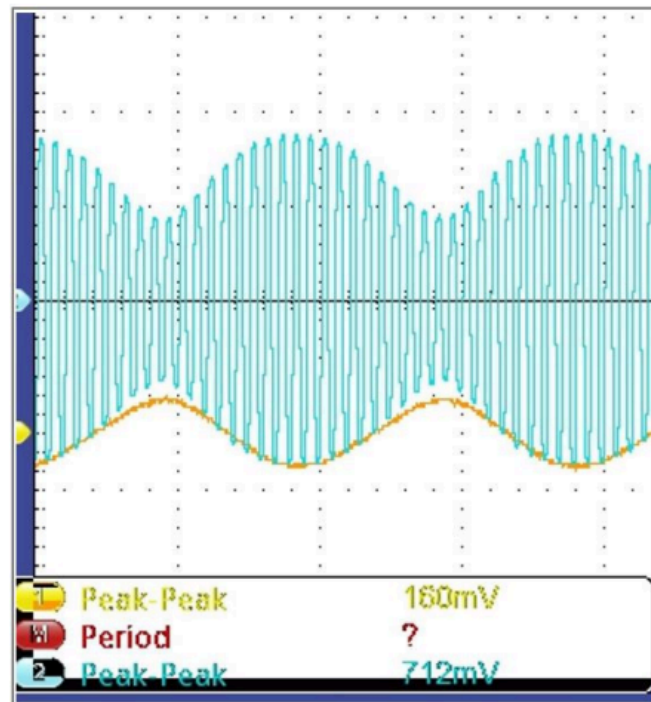


Figure 2: AM Modulated Output(Under modulation) Expected

- Critical / Perfect Modulation ($m = 1$)

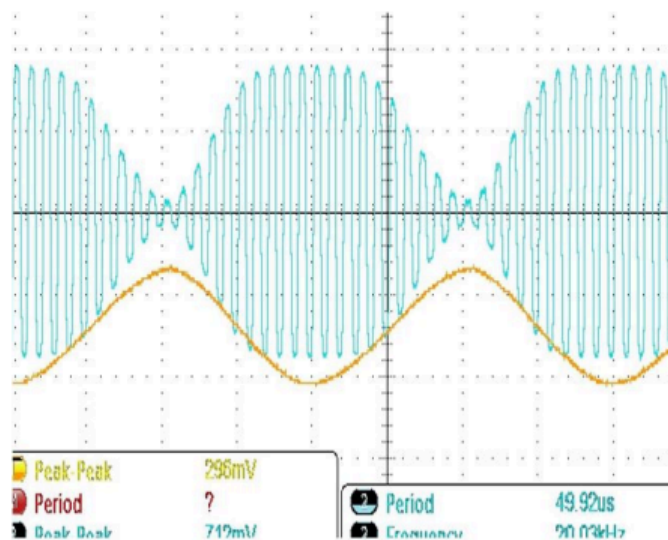


Figure 3: AM Modulated Output(Critical) Expected

- Over Modulation ($m > 1$)

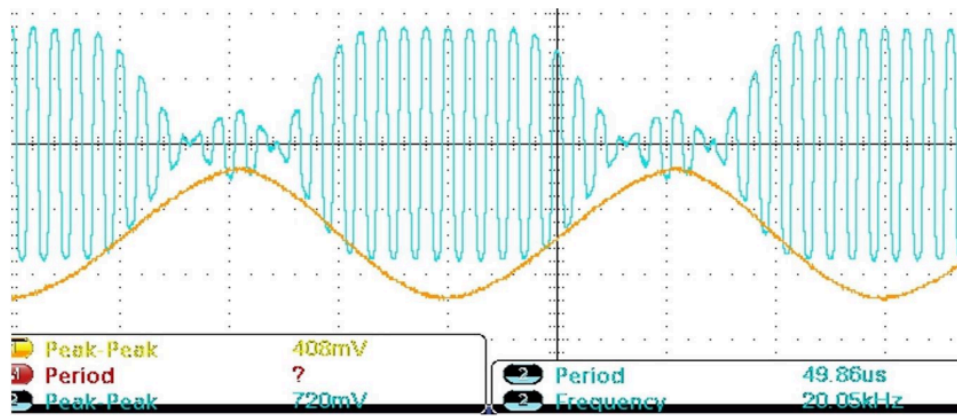


Figure 4: AM Modulated output (Over-modulation) Expected

Task 4: Adjust the potentiometer (VR) to suppress the carrier in the modulated output and observe the Double Sideband Suppressed Carrier (DSB-SC) balanced modulated output, and analyse the signal in the frequency domain.

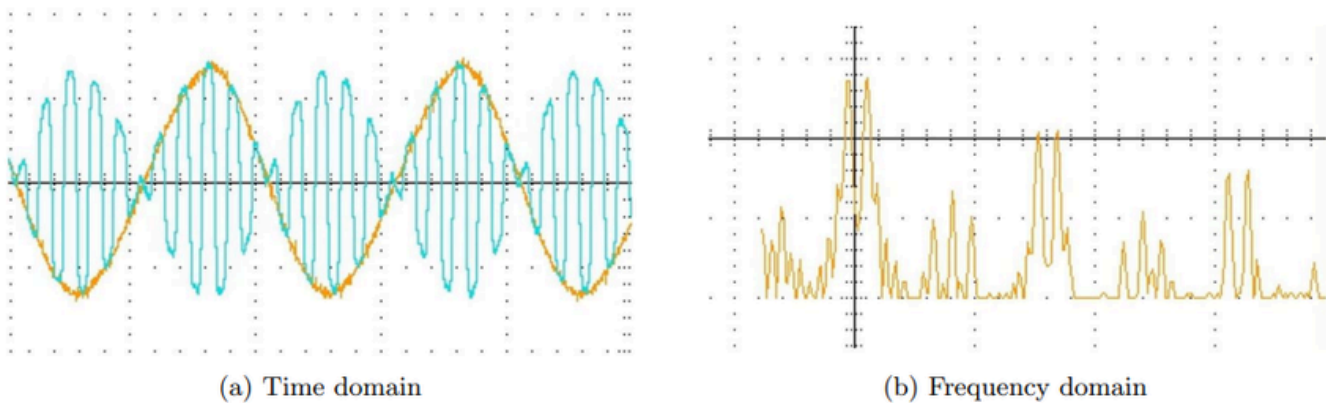


Figure 5: Balanced modulated output Expected

3. Materials Required

- Resistors - 4.7 k Ω (5 units), 3.3 k Ω (2 units), 1 k Ω (3 units), 1 k Ω (3 unit), 18 k Ω (2 units), 100 k Ω (1 unit), 820 Ω (1 unit).
- Capacitors - 0.1 μ F (3 units), 0.47 μ F (1 unit), 22 μ F (1 unit).
- IC MC 1496 (Balanced Modulators/ Demodulator)
- Variable Resistor / Potentiometer
- Signal generator
- Digital Storage Oscilloscope (DSO)
- Power supply for Op-Amps (+5 / -5 / +12 / -12 / GND)
- Breadboard and connecting wires
- Multimeter

4. Theory

Amplitude Modulation (AM)

Amplitude Modulation (AM) is a modulation technique in which the amplitude of a high-frequency carrier signal varies proportionally with the instantaneous amplitude of the message signal, while the carrier frequency and phase remain constant.

Let the modulating signal be

$$m(t) = A_m \cos(2\pi f_m t)$$

and the carrier signal be

$$c(t) = A_c \cos(2\pi f_c t)$$

where

A_m = amplitude of modulating signal

A_c = amplitude of carrier signal

f_m = modulating frequency

f_c = carrier frequency

The AM wave is given by

$$s(t) = [A_c + A_m \cos(2\pi f_m t)] \cos(2\pi f_c t)$$

or

$$s(t) = A_c [1 + m \cos(2\pi f_m t)] \cos(2\pi f_c t)$$

where $m = A_m / A_c$ is called the modulation index, which indicates the degree of modulation.

The modulation index can also be calculated from the envelope of the modulated waveform as

$$m = (A_{\max} - A_{\min}) / (A_{\max} + A_{\min})$$

where

$A_{\max}$ = maximum envelope amplitude

$A_{\min}$ = minimum envelope amplitude

Based on the value of m , three types of modulation occur:

- Under Modulation ($m < 1$) – carrier amplitude varies but envelope does not touch zero.
- Critical Modulation ($m = 1$) – envelope just touches zero, representing 100% modulation.
- Over Modulation ($m > 1$) – distortion occurs because the envelope crosses zero.

Double Sideband Suppressed Carrier (DSB-SC)

In conventional AM, the transmitted signal contains the carrier and two sidebands (USB and LSB). However, the carrier does not carry information and consumes a large portion of the transmitted power.

To improve power efficiency, the carrier component is suppressed and only the sidebands are transmitted. This technique is known as Double Sideband Suppressed Carrier (DSB-SC) modulation.

The DSB-SC signal is obtained by multiplying the carrier and modulating signals.

$$m(t) = A_m \cos(2\pi f_m t)$$

$$c(t) = A_c \cos(2\pi f_c t)$$

DSB-SC signal:

$$s(t) = m(t) \cdot c(t)$$

$$s(t) = A_m A_c \cos(2\pi f_m t) \cos(2\pi f_c t)$$

The spectrum of DSB-SC contains two sidebands at frequencies $(f_c + f_m)$ and $(f_c - f_m)$ and the carrier component is ideally absent.

Bandwidth of DSBSC Wave

$$\text{Bandwidth (BW)} : f_{max} - f_{min} = 2f_m$$

$$f_{max} : f_{max} = f_c + f_m$$

$$f_{min} : f_{min} = f_c - f_m$$

Power Calculations of DSBSC Wave

$$s(t) = \frac{A_m A_c}{2} \cos(2\pi(f_c + f_m)t) + \frac{A_m A_c}{2} \cos(2\pi(f_c - f_m)t)$$

Power of cos signal is given by

$$P = \frac{v_{rms}^2}{R} = \frac{(v_m \sqrt{2})^2}{R}$$

Upper sideband power

$$P_{USB} = \frac{A_m^2 A_c^2}{8R}$$

Lower sideband power

$$P_{LSB} = \frac{A_m^2 A_c^2}{8R}$$

Total Power is given by

$$P_t = P_{LSB} + P_{USB} = \frac{A_m^2 A_c^2}{4R}$$

The total transmitted power is the sum of powers in the upper and lower sidebands.

Balanced Modulator

A balanced modulator is a circuit used to generate Double Sideband Suppressed Carrier (DSB-SC) signals by multiplying the carrier signal with the modulating signal while suppressing the carrier component at the output.

It consists of two identical AM modulators arranged in a balanced configuration. The outputs of these modulators are combined such that the carrier components cancel each other, leaving only the upper and lower sidebands in the output signal.

The balanced modulator improves power efficiency, since the carrier does not carry information and is therefore removed from transmission.

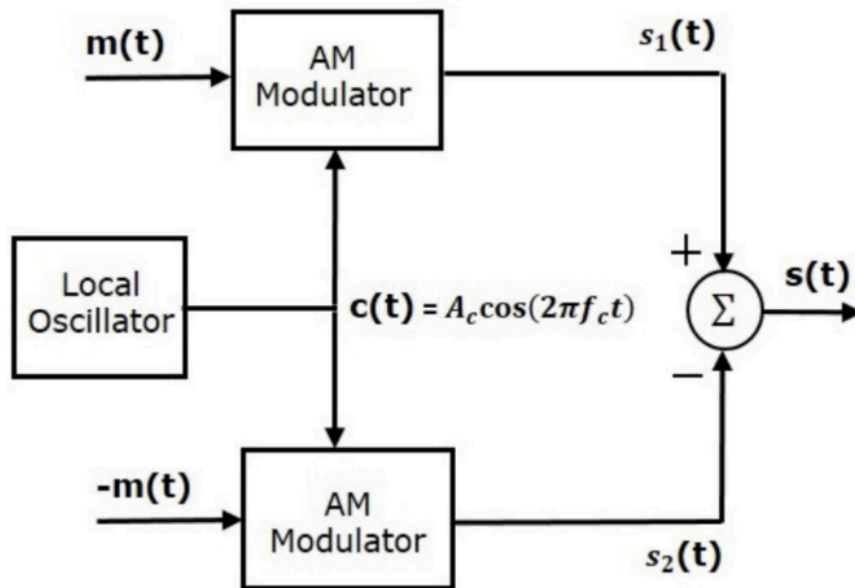


Figure 6: Block Diagram of Balanced Modulator

The output of the balanced modulator corresponds to the standard DSB-SC signal

$$s(t) = A_m A_c \cos(2\pi f_m t) \cos(2\pi f_c t)$$

where the output contains only the two sidebands at $(f_c + f_m)$ and $(f_c - f_m)$ without the carrier component.

IC MC1496

The MC1496 is an integrated circuit commonly used as a balanced modulator or demodulator. It contains differential amplifier stages that perform multiplication of the carrier and modulating signals.

It is capable of generating both AM and DSB-SC signals with low distortion, making it suitable for communication experiments.

Key pins of the IC include:

- **Output Pin** – provides the modulated or demodulated output signal
- **Modulating Signal Input (AM Input)** – input for the message signal
- **Carrier Input (Local Oscillator Input)** – input for the carrier signal
- **VCC** – power supply for the IC
- **GND** – ground connection

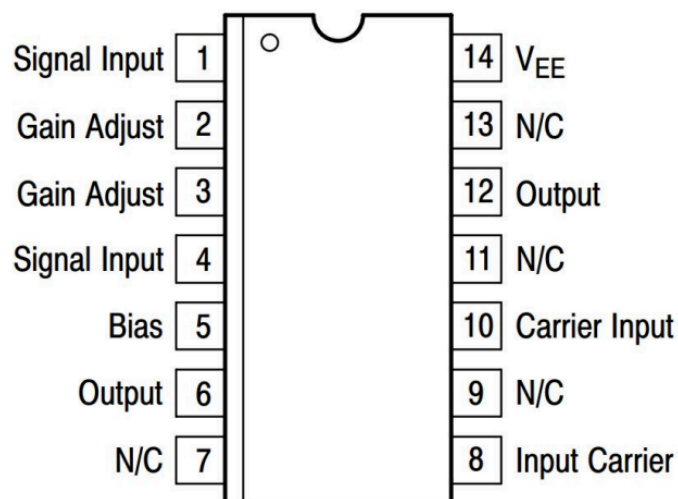


Figure 7: MC 1496 Pin details

Carrier Feedthrough and Carrier Suppression

In practical balanced modulators, a small portion of the carrier may still appear at the output. This unwanted carrier component is called carrier feedthrough.

To minimize this effect, a bias trim potentiometer is adjusted to balance the differential amplifier currents, which results in carrier suppression.

Carrier suppression depends on proper carrier amplitude, circuit layout, and correct signal levels. When the circuit is properly balanced, the output primarily contains only the upper and lower sidebands, producing a clean DSB-SC signal.

5. RESULTS AND OBSERVATIONS

Setting $A_c = 300\text{mV pp}$, $f_c = 20\text{kHz}$ and $A_m = 150\text{mV pp}$, $f_m = 2\text{kHz}$ theoretical calculation of Modulation Index m gives:

$$m = \frac{A_m}{A_c} = \frac{150}{300} = 0.5$$

$$m = \frac{A_m}{A_c} = \frac{A_{max} - A_{min}}{A_{max} + A_{min}}$$

$A_c = 300\text{mV}_{pp}$, $f_c = 20\text{kHz}$ and $A_m = 150\text{mV}_{pp}$, $f_m = 2\text{kHz}$			
Serial Number	$A_{max}(\text{mv})$	$A_{min}(\text{mv})$	Modulation Index
1	317.0	129.40	0.420
2	296.0	136.00	0.370
3	301.0	137.40	0.373
4	300	100	0.50

The resultant waveform for the case when $m = 0.42$:

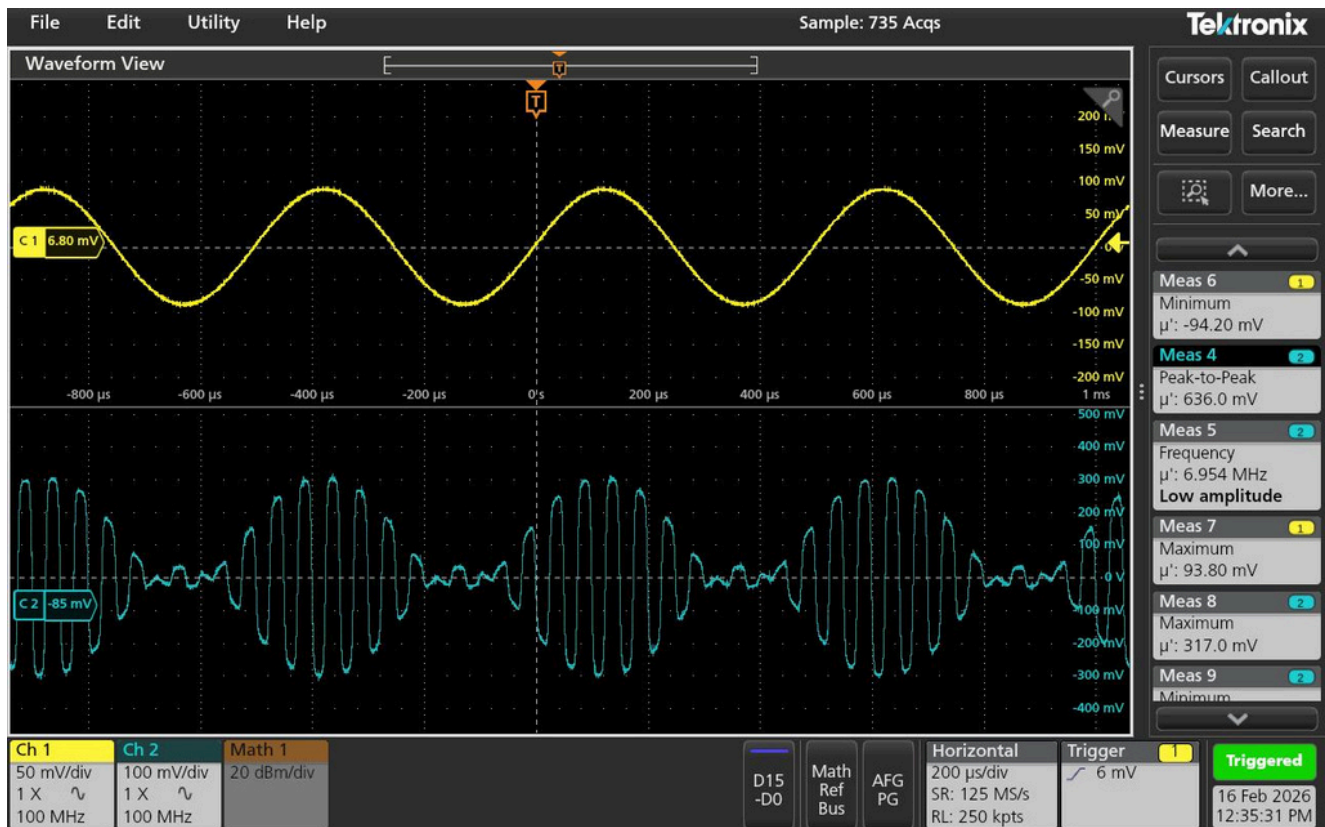


Figure 8: Output Waveform for $m = 0.42$ (Undermodulated)

Under Modulation

For the case of Under Modulation, we set our Inputs as: $A_c = 300\text{mV pp}$, $f_c = 20\text{kHz}$ and $A_m = 150\text{mV pp}$, $f_m = 2\text{kHz}$.

Theoretical calculation of Modulation Index m gives:

$$m = \frac{A_m}{A_c} = \frac{150}{300} = 0.5$$

Since $m = 0.5 < 1$, thus this is Under modulation.

Practically Viewed case for $m < 1$, (Under modulation) were the same as the previous table, two such results close to $m = 0.5$ were:

- Case 1: $A_{\max} = 300\text{mV pp}$, $A_{\min} = 100\text{mV pp}$:

$$m = \frac{A_{\max} - A_{\min}}{A_{\max} + A_{\min}} = \frac{300 - 120}{300 + 120} = 0.42$$

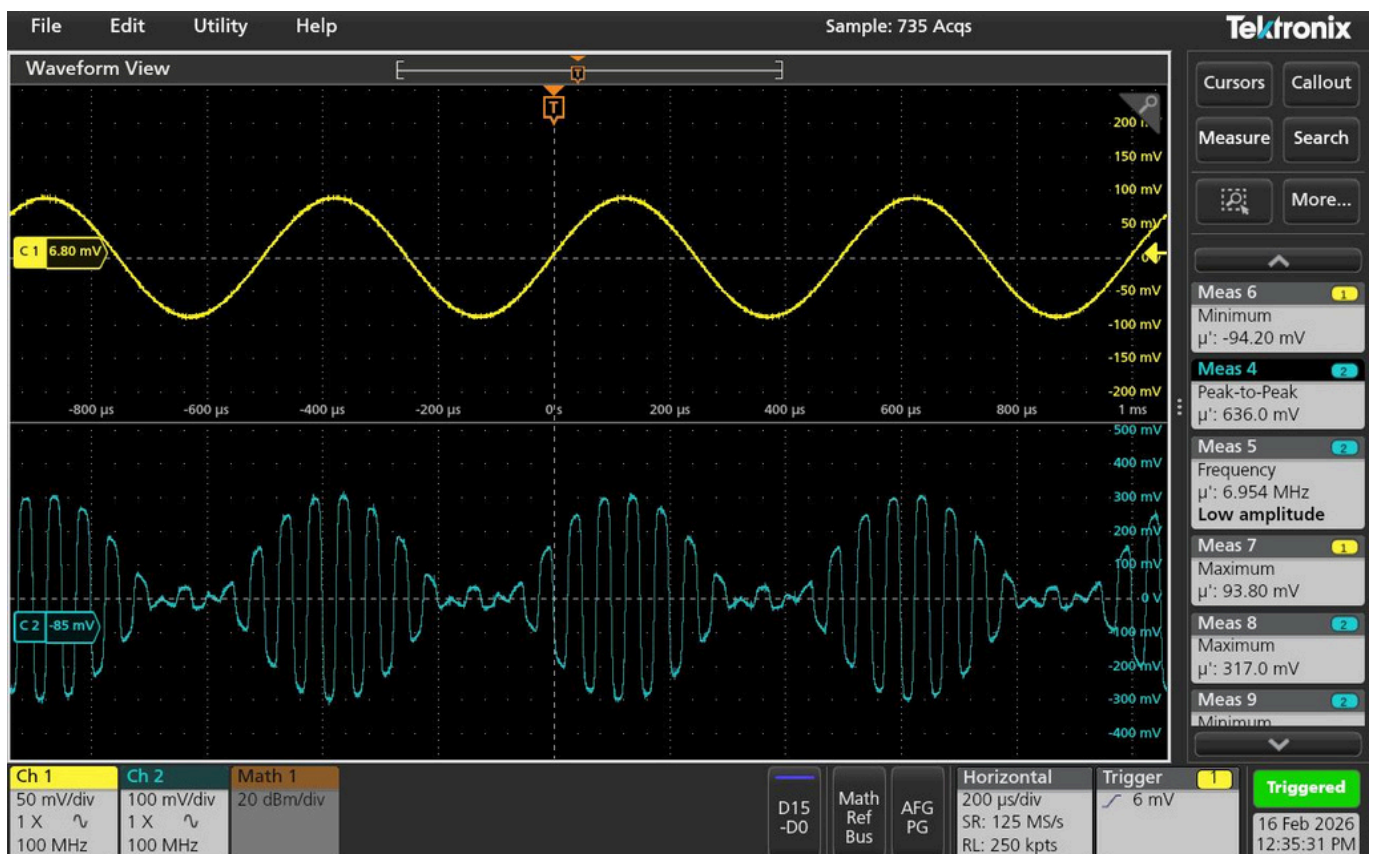


Figure 9: Output Waveform for $m = 0.42$ (Undermodulated)

- Case 2: $A_{\max} = 300\text{mV pp}$, $A_{\min} = 120\text{mV pp}$:

$$m = \frac{A_{max} - A_{min}}{A_{max} + A_{min}} = m = \frac{301.0 - 137.4}{301.0 + 137.4} = 0.373$$

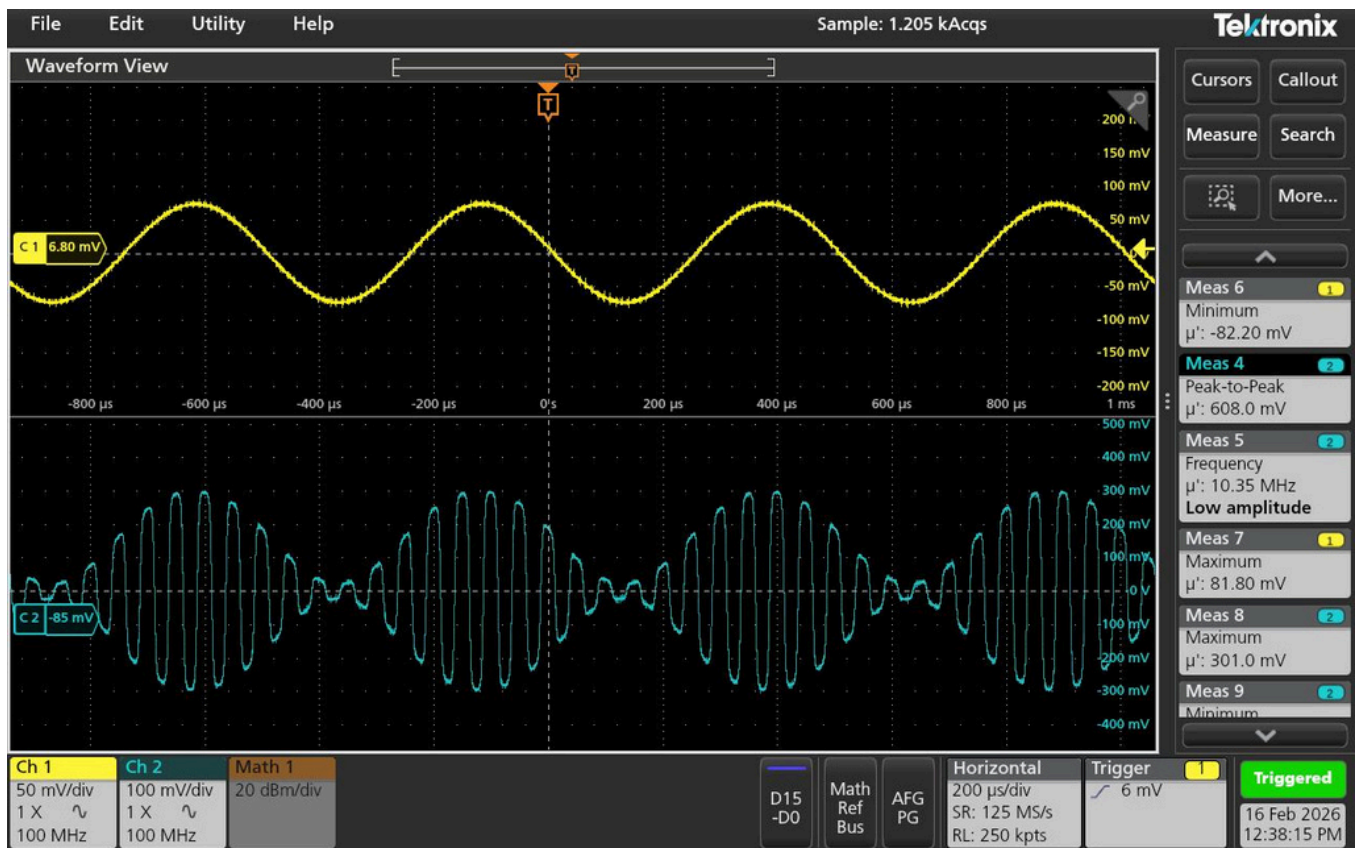


Figure 10: Output Waveform for $m = 0.373$ (Undermodulated)

We observe in the frequency domain diagram 3 peaks, which include the f_c , $f_c + f_m$, $f_c - f_m$ as all present in the frequency spectrum. With the central frequency with much higher power than the sidebands.



Figure 11: FFT for $m = 0.373$ (Undermodulated)

Critical Modulation

For the case of Critical Modulation, we set our Inputs as: $A_c = 300\text{mV pp}$, $f_c = 20\text{kHz}$ and $A_m = 150\text{mV pp}$, $f_m = 2\text{kHz}$.

Theoretical calculation of Modulation Index m gives:

$$m = \frac{A_m}{A_c} = \frac{300}{300} = 1$$

Since $m = 1$, thus this is Critical modulation.

Practically Viewed case for $m = 0.923$, (Under modulation) was $A_{\max} = 294.0\text{mV pp}$, $A_{\min} = 11.80\text{mV pp}$:

$$m = \frac{A_{\max} - A_{\min}}{A_{\max} + A_{\min}} = \frac{294.0 - 11.8}{294.0 + 11.8} = 0.923$$



Figure 12: Output Waveform for $m = 0.923$ (Critically Modulated)

Over Modulation

For the case of Over Modulation, we set our Inputs as: $A_c = 300\text{mV pp}$, $f_c = 20\text{kHz}$ and $A_m = 150\text{mV pp}$, $f_m = 2\text{kHz}$.

Theoretical calculation of Modulation Index m gives:

$$m = \frac{A_m}{A_c} = \frac{450}{300} = 1.5$$

Since $m = 1.5 > 1$, thus this is Over modulation.

Practically Viewed case for $m > 1$, (Under modulation) was $A_{\max} = 300\text{mV pp}$, $A_{\min} = -100\text{mV pp}$

$$m = \frac{A_{\max} - A_{\min}}{A_{\max} + A_{\min}} = \frac{289.0 - (-18.60)}{289.0 + (-18.60)} = 1.138$$

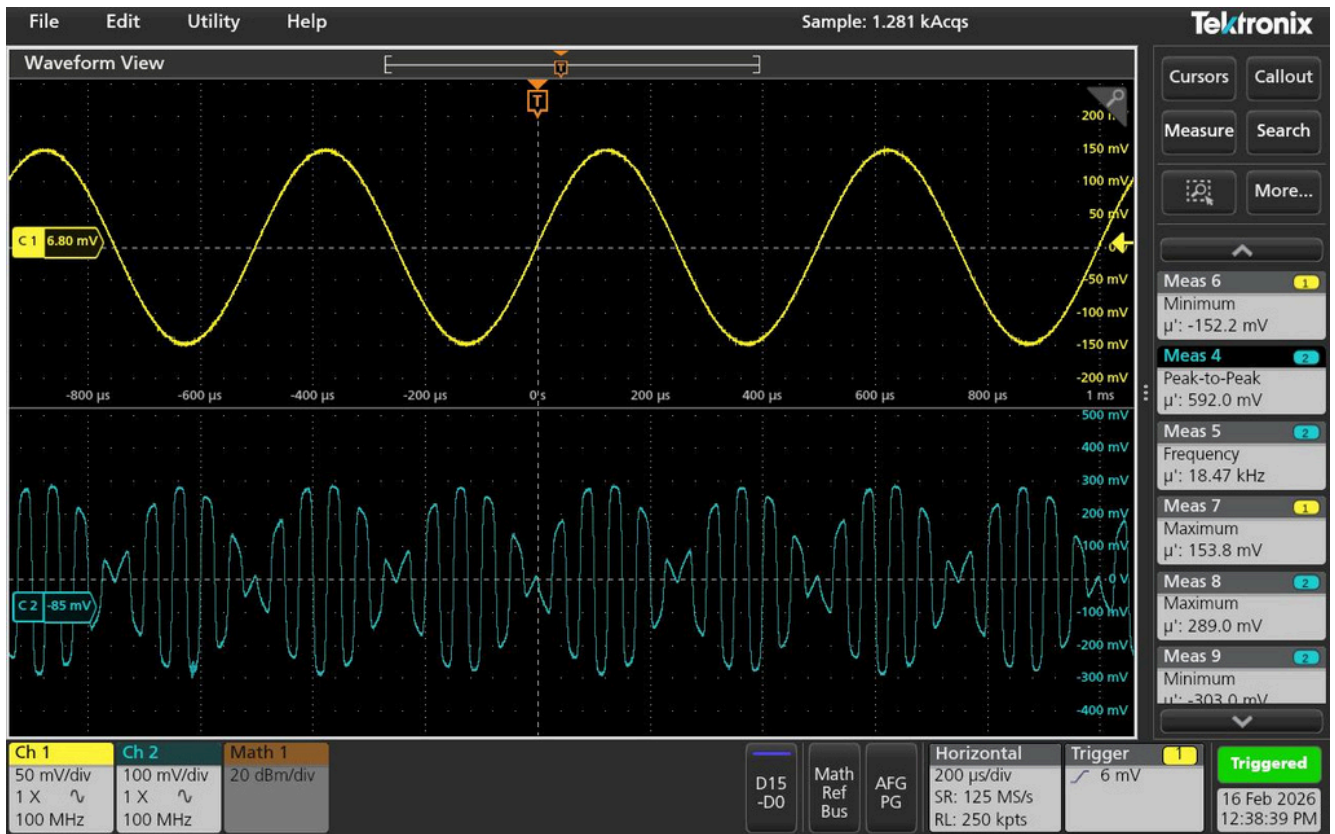


Figure 13: Output Waveform for $m = 1.138$ (Over Modulated)

DSB-SC

We changed the potentiometer value to attain a stage such that the carrier component frequency turned close to zero:

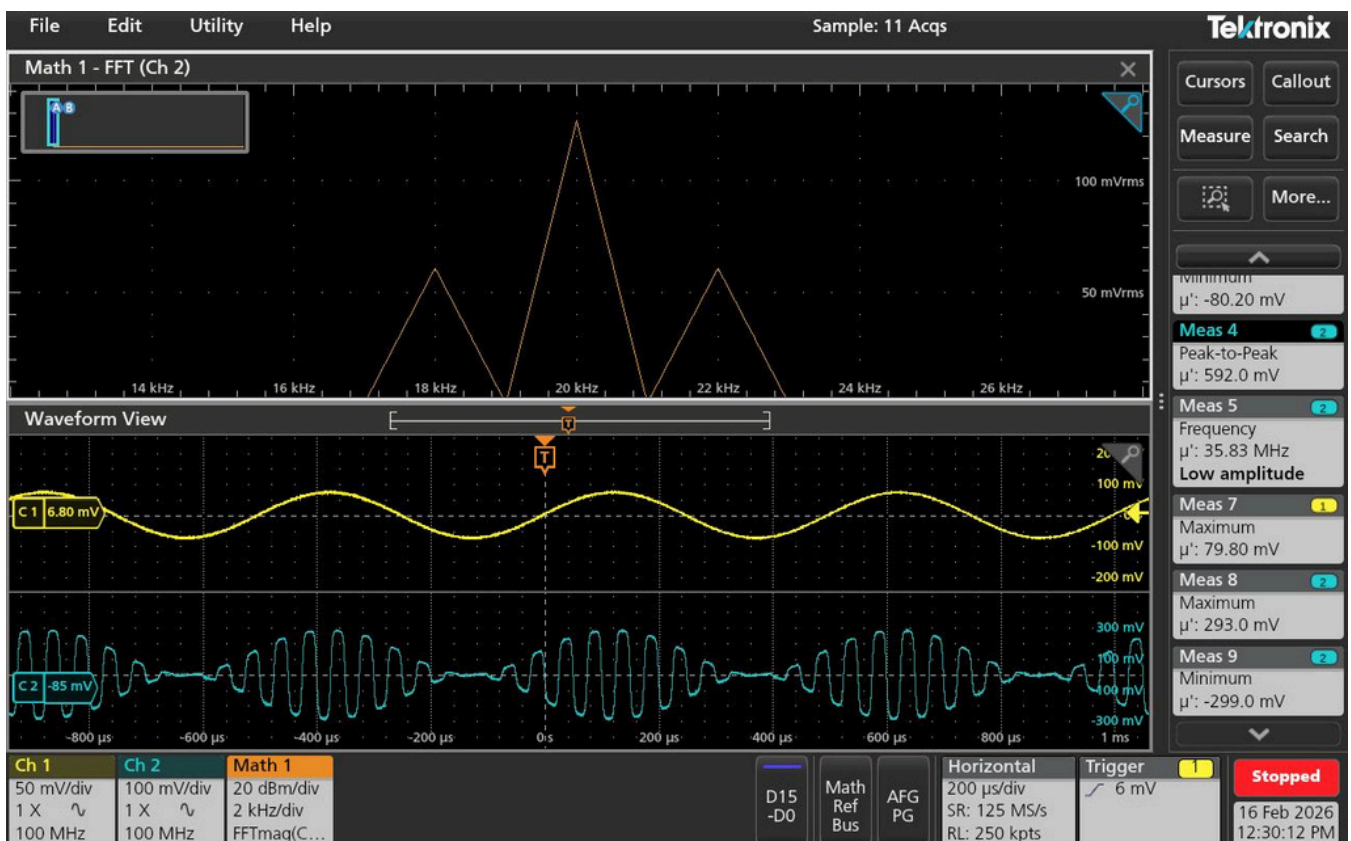


Figure 14: DSB-SC from Over-Modulated Case



Figure 15: DSB-SC from Critically-Modulated Case

In the FFT diagram, only three frequencies have power distributed since the carrier has been suppressed.

6. Discussion

- The MC1496 is a balanced modulator–demodulator IC widely used in analog communication systems for implementing Amplitude Modulation (AM) and Double Sideband Suppressed Carrier (DSB-SC) modulation. Internally, it consists of differential amplifier stages formed using transistors, which enable multiplication of the carrier and message signals.
- In the experiment, the carrier signal and modulating signal are applied to different input ports of the IC, and the modulated output is obtained at the output pin. The gain and carrier suppression level can be adjusted using potentiometers provided in the circuit, typically called gain adjust and balance adjust.
- The modulation index (m) indicates the depth of modulation and can be determined in two ways:

- From the input signals:

$$m = A_m / A_c$$

- From the output waveform envelope:

$$m = (A_{\max} - A_{\min}) / (A_{\max} + A_{\min})$$

- Ideally, both methods give the same value. However, in practical circuits, noise, component tolerances, distortion, and non-linearities in the amplifier stages may introduce small deviations.
- When the carrier and message signals are multiplied inside the modulator, the output signal contains three frequency components:
 f_c (carrier), $f_c + f_m$ (upper sideband), and $f_c - f_m$ (lower sideband).
 Using trigonometric identities, it can be shown that the AM signal is composed of these spectral components.
- By adjusting the balance potentiometer, the carrier component can be minimized or suppressed. When the carrier is ideally removed, the output contains only the two sidebands, resulting in a Double Sideband Suppressed Carrier (DSB-SC) signal.
- The balanced modulator operates using two symmetrical differential amplifier paths. When both paths are perfectly balanced, the carrier components cancel each other, leading to carrier suppression. If the balance is slightly disturbed, a DC offset appears, which introduces a portion of the carrier in the output, resulting in standard AM output.
- During the experiment, different modulation conditions were observed by varying the amplitude of the message signal:
 - Under-modulation ($m < 1$) – envelope does not reach zero.
 - Critical modulation ($m = 1$) – envelope just touches zero (100% modulation).
 - Over-modulation ($m > 1$) – envelope distortion occurs and proper demodulation becomes difficult.
- In the frequency domain, the spectrum clearly shows the upper and lower sidebands around the carrier frequency. In the DSB-SC case, the carrier peak is significantly reduced, demonstrating successful carrier suppression.
- One advantage of DSB-SC modulation is improved power efficiency, since the carrier, which does not carry useful information, is suppressed and the transmitted power is concentrated in the sidebands.
- However, DSB-SC systems require coherent detection at the receiver, meaning the receiver must generate a carrier signal with the same frequency and phase as the original carrier. Any phase or frequency mismatch can lead to distortion in the recovered message signal.
- Experimental observations confirmed that the MC1496 IC can function both as an AM modulator and as a balanced modulator, depending on the adjustment of the carrier balance potentiometer.